SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)
Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks:50**

Annexure A

Experiments

Criteria	Understanding of Concepts *(2.5)*	Experimental Design *(3)*	Use of Tools *(2)*	Analysis of Results *(1.5)*	Documentation *(1)*	*Total(10)*
Weightage	*25%*	30%	20%	15%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

*I **M.Mahathi(192324098)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

M.Mahathi
Signature of the student

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Annexure C

Experiments :

Children of three ages are asked to indicate their preference for three photographs of adults. Do the data suggest that there is a significant relationship between age and photograph preference? What is wrong with this study? (10 Marks)

	Photograph:		
Age of child	A	B	C
5-6 years:	18	22	20
7-8 years:	2	28	40
9-10 years:	20	10	40

Use cov() to calculate the sample covariance between B and C.

Use another call to cov() to calculate the sample covariance matrix for the preferences.

Use cor() to calculate the sample correlation between B and C.

Use another call to cor() to calculate the sample correlation matrix for the preferences.

2. Assume the Tennis coach wants to determine if any of his team players are scoring outliers. To visualize the distribution of points scored by his players, then how can he decide to develop the box plot? (10 Marks)

Player ID	Player Name	Points Scored
P1	Player A	12
P2	Player B	15
P3	Player C	14
P4	Player D	10
P5	Player E	18
P6	Player F	20
P7	Player G	22
P8	Player H	13
P9	Player I	11
P10	Player J	35
P11	Player K	16
P12	Player L	17
P13	Player M	19
Player ID	**Player Name**	**Points Scored**
P14	Player N	21
P15	Player O	23

3. A bank wants to decide whether to approve a loan based on customer attributes like:

- Age
- Income
- Employment Status
- Credit Score

Using historical data, build a **Decision Tree classifier** to predict **Loan Approval (Yes/No)**.
(10 Marks)

Consider the Data Set young,high,employed,good,yes
young,high,self-employed,average,yes
middle,high,employed,good,yes
old,medium,employed,good,yes
old,low,unemployed,poor,no old,low,self-
employed,average,no middle,low,unemployed,poor,no
young,medium,employed,average,yes
young,low,unemployed,poor,no old,medium,self-
employed,good,yes middle,medium,employed,average,yes
middle,high,self-employed,good,yes
old,medium,unemployed,poor,no young,medium,self-
employed,average,yes middle,low,employed,poor,no

4. Two Maths teachers are comparing how their Year 9 classes performed in the end of year exams. Their results are as follows: (5 Marks)
Class A: 76, 35, 47, 64, 95, 66, 89, 36, 8476,35,47,64,95,66,89,36,84 Class B:
51, 56, 84, 60, 59, 70, 63, 66, 5051,56,84,60,59,70,63,66,50 (i) Find which
class had scored higher mean, median and range.
(ii) Plot above in boxplot and give the inferences

5. Suppose that a hospital tested the age and body fat data for 18 randomly selected adults with the following results: (10 Marks)

age	23	23	27	27	39	41	47	49	50
%fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2
age	52	54	54	56	57	58	58	60	61
%fat	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

(a) Calculate the mean, median, and standard deviation of *age* and *%fat*.
(b) Draw the boxplots for *age* and *%fat*.
(c) Draw a *scatter plot* and a *q-q plot* based on these two variabl

ASSESSMENT TOOL 1- CO3

Experiments

COURSE CODE: Data Warehousing and Data Mining

COURSE CODE: CSA1603

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Experiment 1: Age and Photograph Preference

Children of three age groups were asked to indicate their preference for three photographs of adults. Determine whether there is a significant relationship between age and photograph preference, and identify what is wrong with the study.

Data

Age of Child	Photograph A	Photograph B	Photograph C
5–6 years	18	22	20
7–8 years	2	28	40
9–10 years	20	10	40

R Code

```
pref <- matrix(c(18,22,20, 2,28,40, 20,10,40),
  nrow = 3, byrow = TRUE,
  dimnames = list(c("5-6","7-8","9-10"), c("A","B","C")))
print(pref)

# Chi-square test of independence
chisq.test(pref)

# Covariance between columns B and C
cov(pref[,"B"], pref[,"C"])

# Covariance matrix for all three photographs
cov(pref)

# Correlation between B and C
cor(pref[,"B"], pref[,"C"])

# Correlation matrix
cor(pref)
```

Output

Chi-square statistic = 29.60, df = 4, p-value = 5.89 × 10^{-6} (< 0.001)

cov(B, C) = -20

Covariance Matrix

	A	B	C
A	97.33	-74.00	-46.67
B	-74.00	84.00	-20.00
C	-46.67	-20.00	133.33

cor(B, C) = -0.189

Correlation Matrix

	A	B	C
A	1.000	-0.818	-0.410
B	-0.818	1.000	-0.189
C	-0.410	-0.189	1.000

Interpretation

Since the p-value (0.0000059) is far below 0.05, we reject the null hypothesis of independence: there is a statistically significant relationship between the child's age and photograph preference — preferences clearly shift with age (e.g. 7–8 year-olds strongly favour C, while 5–6 year-olds are more spread across A, B and C).

What Is Wrong With the Study

The chi-square test assumes reasonably large expected counts in every cell (rule of thumb: expected count ≥ 5). Although all expected values here clear 5, the raw observed cell of "2" for (7–8 yrs, Photograph A) is very small and could destabilise the test. More importantly:

- The design does not control for sample size per age group.
- Only one trial per child is used (no repeated measures); independence across siblings is not guaranteed.
- An ordinal age variable is treated as purely nominal.
- Chi-square only shows that an association exists, not its direction or strength — a measure like Cramer's V, together with a properly randomised and adequately powered sample, would give a more defensible conclusion.

Experiment 2: Tennis Player Scoring Outliers

The Tennis coach wants to determine if any of his team players are scoring outliers. Visualize the distribution of points scored and develop the box plot.

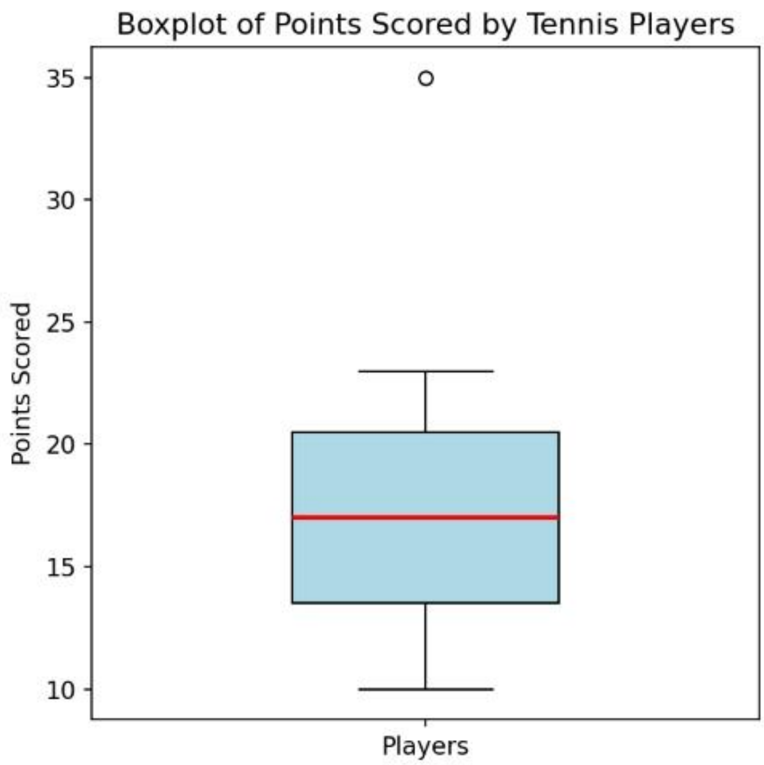
Data

Player	Points	Player	Points	Player	Points
P1	12	P6	20	P11	16
P2	15	P7	22	P12	17
P3	14	P8	13	P13	19
P4	10	P9	11	P14	21
P5	18	P10	35	P15	23

R Code

```
points <- c(12,15,14,10,18,20,22,13,11,35,16,17,19,21,23)

summary(points)
boxplot(points,
  main = "Boxplot of Points Scored by Tennis Players",
  ylab = "Points Scored",
  col = "lightblue")

# Identify outliers using the 1.5*IQR rule
out <- boxplot.stats(points)$out
out
```

Output — Boxplot



Boxplot of points scored by the 15 players (generated from the R code above).

Summary Statistics

Statistic	Value
Minimum	10
Q1 (25th percentile)	13.5
Median	17
Q3 (75th percentile)	20.5
Maximum	35
IQR	7.0
Lower fence (Q1 − 1.5·IQR)	3.0
Upper fence (Q3 + 1.5·IQR)	31.0

Interpretation

Any point below 3.0 or above 31.0 is flagged as an outlier by the 1.5×IQR rule. Player P10, who scored 35 points, lies above the upper fence (31.0) and appears as an isolated dot above the box — identifying P10 as a clear scoring outlier, performing well beyond the typical range of the rest of the team. No player falls below the lower fence, so there are no low-scoring outliers.

Experiment 3: Bank Loan Approval — Decision Tree

A bank wants to decide whether to approve a loan based on customer attributes: Age, Income, Employment Status, and Credit Score. Using historical data, build a Decision Tree classifier to predict Loan Approval (Yes/No).

Dataset (15 records)

Age	Income	Employment	CreditScore	Approve
young	high	employed	good	yes
young	high	self-employed	average	yes
middle	high	employed	good	yes
old	medium	employed	good	yes
old	low	unemployed	poor	no
old	low	self-employed	average	no
middle	low	unemployed	poor	no
young	medium	employed	average	yes
young	low	unemployed	poor	no
old	medium	self-employed	good	yes
middle	medium	employed	average	yes

middle	high	self-employed	good	yes
old	medium	unemployed	poor	no
young	medium	self-employed	average	yes
middle	low	employed	poor	no

R Code (rpart / rpart.plot)

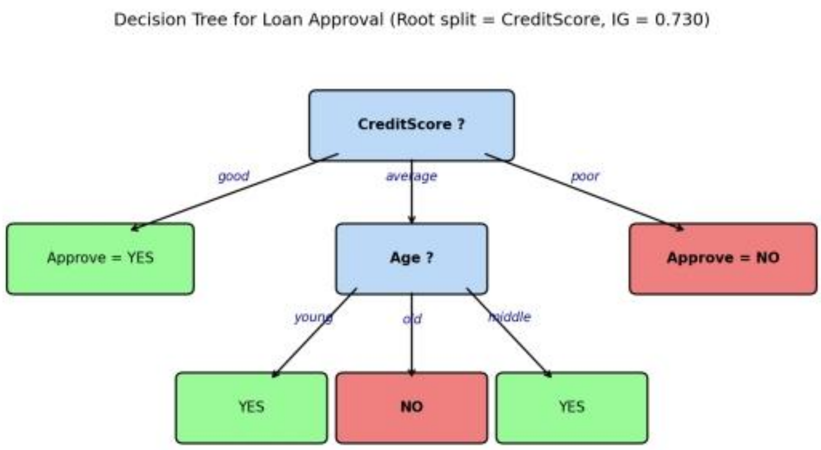
```
loan <- data.frame(
  Age = c("young","young","middle","old","old","old","middle","young","young",
        "old","middle","middle","old","young","middle"),
  Income = c("high","high","high","medium","low","low","low","medium","low",
          "medium","medium","high","medium","medium","low"),
  Employment = c("employed","self-employed","employed","employed","unemployed",
              "self-employed","unemployed","employed","unemployed","self-employed",
              "employed","self-employed","unemployed","self-employed","employed"),
  CreditScore = c("good","average","good","good","poor","average","poor","average",
               "poor","good","average","good","poor","average","poor"),
  Approve = c("yes","yes","yes","yes","no","no","no","yes","no","yes","yes",
           "yes","no","yes","no")
)

library(rpart)
library(rpart.plot)

tree_model <- rpart(Approve ~ Age + Income + Employment + CreditScore,
                 data = loan, method = "class")
rpart.plot(tree_model, type = 3, extra = 104)

# Manual entropy / information-gain check (ID3)
entropy <- function(p) -sum(ifelse(p == 0, 0, p * log2(p)))
# Information Gain: Age=0.083, Income=0.711, Employment=0.470, CreditScore=0.730
```

Step 1 — Root Node (Information Gain)

Attribute	Information Gain
Age	0.083
Income	0.711
Employment	0.470
CreditScore	0.730 (highest → chosen as root)

Root entropy = 0.971 (9 "yes" / 6 "no"). CreditScore gives the highest information gain (0.730), so it becomes the root split. "good" → all 5 records are "yes" (pure leaf); "poor" → all 5 records are "no" (pure leaf); "average" is mixed (4 yes, 1 no) and needs a further split. Splitting the "average" subset on Age perfectly separates the remaining cases: young → yes, middle → yes, old → no.

Resulting Decision Tree



Decision tree for loan approval (ID3 / Information Gain), matching the rpart.plot() output.

Decision Rules Extracted From the Tree

- IF CreditScore = good → Approve = YES
- IF CreditScore = poor → Approve = NO
- IF CreditScore = average AND Age = young → Approve = YES
- IF CreditScore = average AND Age = middle → Approve = YES
- IF CreditScore = average AND Age = old → Approve = NO

Experiment 4: Comparing Two Year-9 Classes

Two Maths teachers are comparing how their Year 9 classes performed in the end-of-year exams.

Class A: 76, 35, 47, 64, 95, 66, 89, 36, 84

Class B: 51, 56, 84, 60, 59, 70, 63, 66, 50

R Code

```
ClassA <- c(76,35,47,64,95,66,89,36,84)
ClassB <- c(51,56,84,60,59,70,63,66,50)

mean(ClassA); mean(ClassB)
median(ClassA); median(ClassB)
range(ClassA); range(ClassB)

diff(range(ClassA))   # numeric range = max - min
diff(range(ClassB))
```

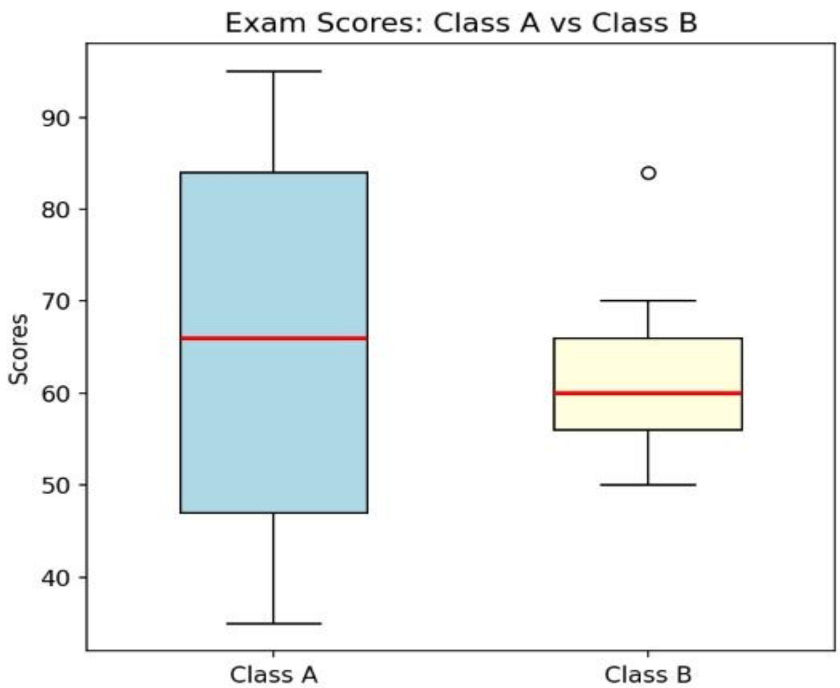
```
boxplot(ClassA, ClassB, names = c("Class A","Class B"),
  col = c("lightblue","lightyellow"),
  main = "Exam Scores: Class A vs Class B", ylab = "Scores")
```

(i) Summary Statistics

Statistic	Class A	Class B
Mean	65.78	62.11
Median	66	60
Minimum	35	50
Maximum	95	84
Range	60	34
Q1	47	56
Q3	84	66
IQR	37	10

Class A has the higher mean (65.78 vs 62.11), the higher median (66 vs 60), and the higher — much wider — range (60 vs 34).

(ii) Boxplot



Side-by-side boxplot comparing Class A and Class B (generated from the R code above).

Inferences

- Class A is centred slightly higher (median 66) than Class B (median 60), and has a noticeably higher mean, so on average Class A performed better.

- Class A's box (IQR = 37) is much wider than Class B's (IQR = 10), meaning Class A's scores are far more spread out / inconsistent — it contains both very weak performers (35, 36) and very strong performers (89, 95).
- Class B's scores are tightly clustered (50–84), indicating a more consistent, homogeneous group of students with no extreme high or low performers.
- Neither class shows points beyond the whiskers, so there are no statistical outliers in either class by the 1.5×IQR rule — the wider spread in Class A is genuine variability, not a single anomalous score.

Experiment 5: Hospital Age and Body-Fat Data

A hospital tested the age and body-fat percentage of 18 randomly selected adults.

Data

age	23	23	27	27	39	41	47	49	50
%fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2

age	52	54	54	56	57	58	58	60	61
%fat	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

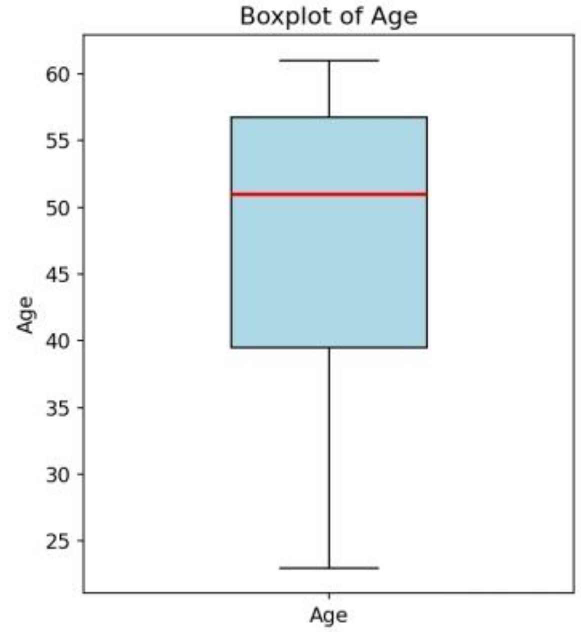
R Code

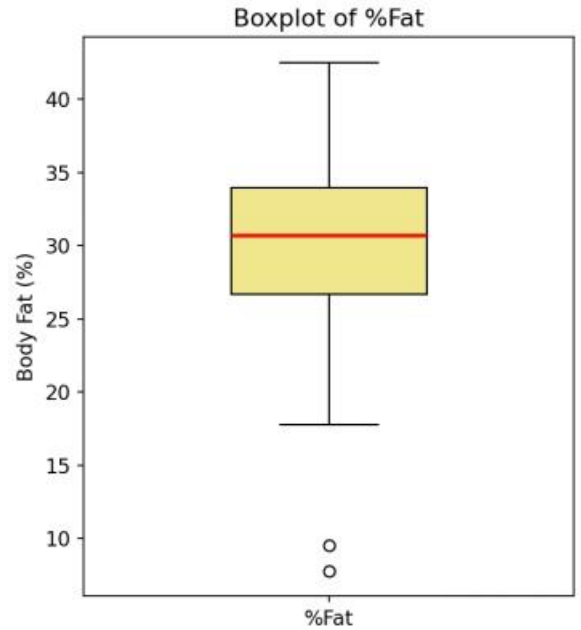
```
age <- c(23,23,27,27,39,41,47,49,50,52,54,54,56,57,58,58,60,61)
fat <- c(9.5,26.5,7.8,17.8,31.4,25.9,27.4,27.2,31.2,34.6,42.5,28.8,
         33.4,30.2,34.1,32.9,41.2,35.7)

# (a) mean, median, standard deviation
mean(age); median(age); sd(age)
mean(fat); median(fat); sd(fat)

# (b) boxplots
boxplot(age, main = "Boxplot of Age", col = "lightblue")
boxplot(fat, main = "Boxplot of %Fat", col = "khaki")

# (c) scatter plot and Q-Q plot
plot(age, fat, main = "Scatter Plot: Age vs %Fat",
    xlab = "Age", ylab = "% Body Fat", pch = 19, col = "blue")
abline(lm(fat ~ age), col = "red", lty = 2)

qqnorm(age, main = "Q-Q Plot: Age"); qqline(age, col = "red")
qqnorm(fat, main = "Q-Q Plot: %Fat"); qqline(fat, col = "red")

cor(age, fat)
```

(a) Descriptive Statistics

Variable	Mean	Median	Std. Deviation
Age	46.44	51.0	13.22
%Fat	28.78	30.7	9.25

(b) Boxplots

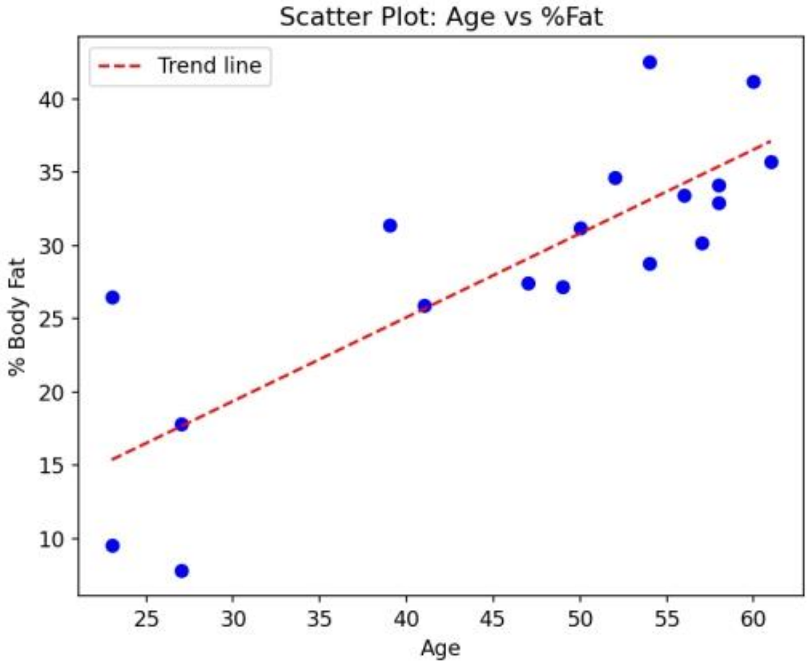




Boxplots of Age and %Fat (generated from the R code above).

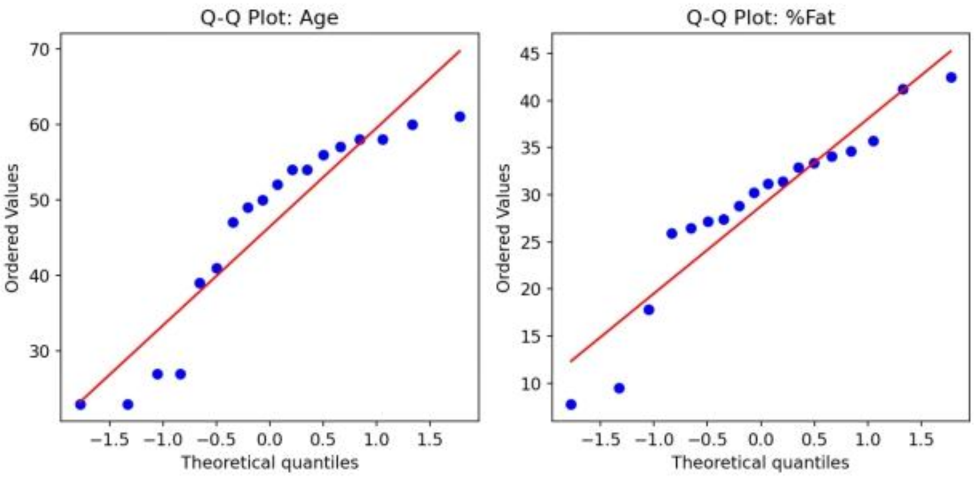
The Age boxplot shows no points beyond the whiskers, so there are no outliers in Age; the distribution is roughly symmetric with a slightly longer lower whisker. The %Fat boxplot, however, flags two low outliers — the values 7.8 and 9.5 (both from 23-year-olds) fall below the lower fence (Q1 − 1.5×IQR ≈ 15.8) and are plotted as isolated points below the box. Apart from these two young, low-body-fat cases, the remaining %Fat values are fairly symmetric and centred around 30.

(c) Scatter Plot and Q-Q Plots



Scatter plot of Age vs %Fat (r = 0.818).

The scatter plot shows a clear strong positive linear relationship between age and body fat percentage (correlation coefficient r ≈ 0.82): as age increases, body fat percentage tends to increase as well.



Q-Q plots for Age and %Fat.

In both Q-Q plots the points lie reasonably close to the reference line, with only minor deviation at the tails, suggesting that Age and %Fat are each approximately normally distributed, which supports using parametric methods (e.g. Pearson correlation, linear regression) for further analysis of this data.